Structure from DBB: Manual

Version 1.0.0 · Henrique Gaspar, NTNU · November 2026 · shiplab.github.io/struct

> **For educational use only.** Structure from DBB is made for teaching and early concept exploration. Its results are simplified estimates; they must not be used for the design, approval or construction of any ship, for class or flag-state documentation, or for any safety-critical decision. Provided as it is, without warranty of any kind; the authors accept no liability for its use.

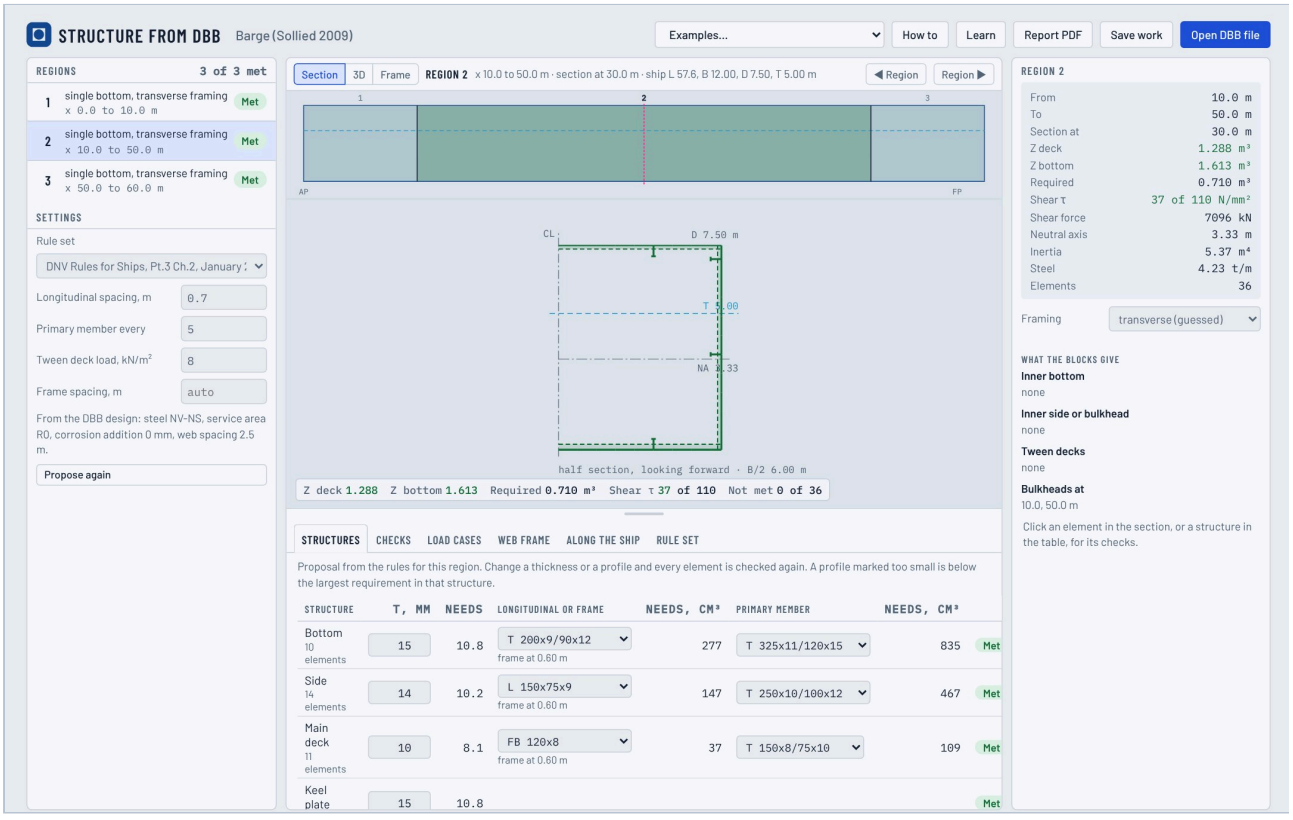


Figure 1. The app as it opens, with the barge of the compendium: regions on the left, the side view and the midship section in the centre, the tables below and the selected region on the right.

1. What Structure from DBB is

Structure from DBB proposes and checks the main hull structure of a ship from its Design Building Block (DBB) arrangement. You open a design saved by the Ship Design Building Block Editor. The app reads the hull and the blocks, splits the ship into regions, and gives you three things:

1. **Longitudinal structure per region.** Plates, longitudinals and primary members are sized to the DNV rules, with the hull girder section modulus checked against the rule value along the length.
2. **Transverse web frame per region.** The frame is analysed as a beam model under load cases taken from the blocks, as the 3D Beam program of Nauticus Hull does. Its girders are sized from the computed forces.
3. **Transverse bulkheads.** They are found in the arrangement and sized with a simplified rule method. A 3D view shows everything on the real hull.

Every value can be changed by hand, and every element is checked again at once: green is met, red is not met. Each check names its rule and shows the required and the achieved value.

Who it is for. Students of ship design and naval architecture, and designers at concept stage who want to see how an arrangement drives the structure. It is for educational use only: a teaching and exploration tool, not a design or class tool (see section 10).

Where it comes from:

- It continues the Hull Scantlings Design Assistant made in 2018 by Olivia Chaves in her master thesis at NTNU, presented at ICSOS 2018. That app was meant as an open, simplified Nauticus Hull.
- Structure from DBB adds what a DBB design knows: the real hull, the regions and the inner boundaries from the blocks, the space contents for the load cases, and a web frame analysis in the manner of 3D Beam.
- Parts of it are based on the compendium *Strukturdesign av skip* by Arne Jan Sollied (Høgskolen i Ålesund, 2009): the barge example and the lessons under Learn, the rule distributions of the moments along the length, the buckling with the Johnson correction, the plating as flange of the web frames and the lightening holes.
- It runs on the core of the DBB editor and is meant to become its Structure tab.

The flow: DBB design (hull and blocks) → transverse bulkheads → regions → for each region its section (longitudinal structure, hull girder), its web frame (load cases, beam model) and its bulkheads → checks, report and 3D view.

The blocks place the bulkheads, and the bulkheads cut the ship into regions. Each region then gets its longitudinal structure, its web frame and its bulkheads checked.

2. Getting started

Open a design and the app makes a full proposal at once, so the first screen already shows a met or not-met result for every region.

1. **Open a design.** Press *Open DBB file* and choose the .json that the DBB editor saves (Save). The design needs exactly one hull; the blocks below the main deck give the layout. Or pick one of seven examples under *Examples*: the barge of the compendium (opened at start), R/V Gunnerus (real hull from the open Gunnerus data), PSV, AHTS, double hull tanker, feeder container ship and handysize bulk carrier (section 3 describes each).
2. **Pick a region** in the side view at the top, in the Regions list, or with the left and right arrow keys.
3. **Choose a view** with the buttons above the drawing: *Section* (half section with every element), *3D* (the hull with the structure of all regions) or *Frame* (the web frame as a beam model).
4. **Read and change** the tables in the drawer under the drawing, and click any element, structure or frame member for its checks in the right panel.

The screen

Area	What it holds
Top bar	Design name, Examples, How to (help and About), Learn (the lessons, section 11), Report PDF, Save work, Open DBB file
Left panel	Regions with their status; Settings: rule set, longitudinal spacing, primary member every n-th longitudinal, tween deck load, Propose again
Centre, top	View switch, region title, view tools (camera views in 3D; load case, Spaces, Loads and diagram in Frame)
Centre, side view	The hull profile with the regions coloured by status, the bulkheads and the section position
Centre, drawing	Section, 3D or Frame, with a readout of the main results at its foot
Drawer tabs	Structures, Checks, Load cases, Web frame, Along the ship, Rule set. Drag the bar above them, or the row of tab names, to make them taller (the drawing gives way); double-click for the usual height
Right panel	The selected region, element or frame member: values, the blocks used and the log of checked rules

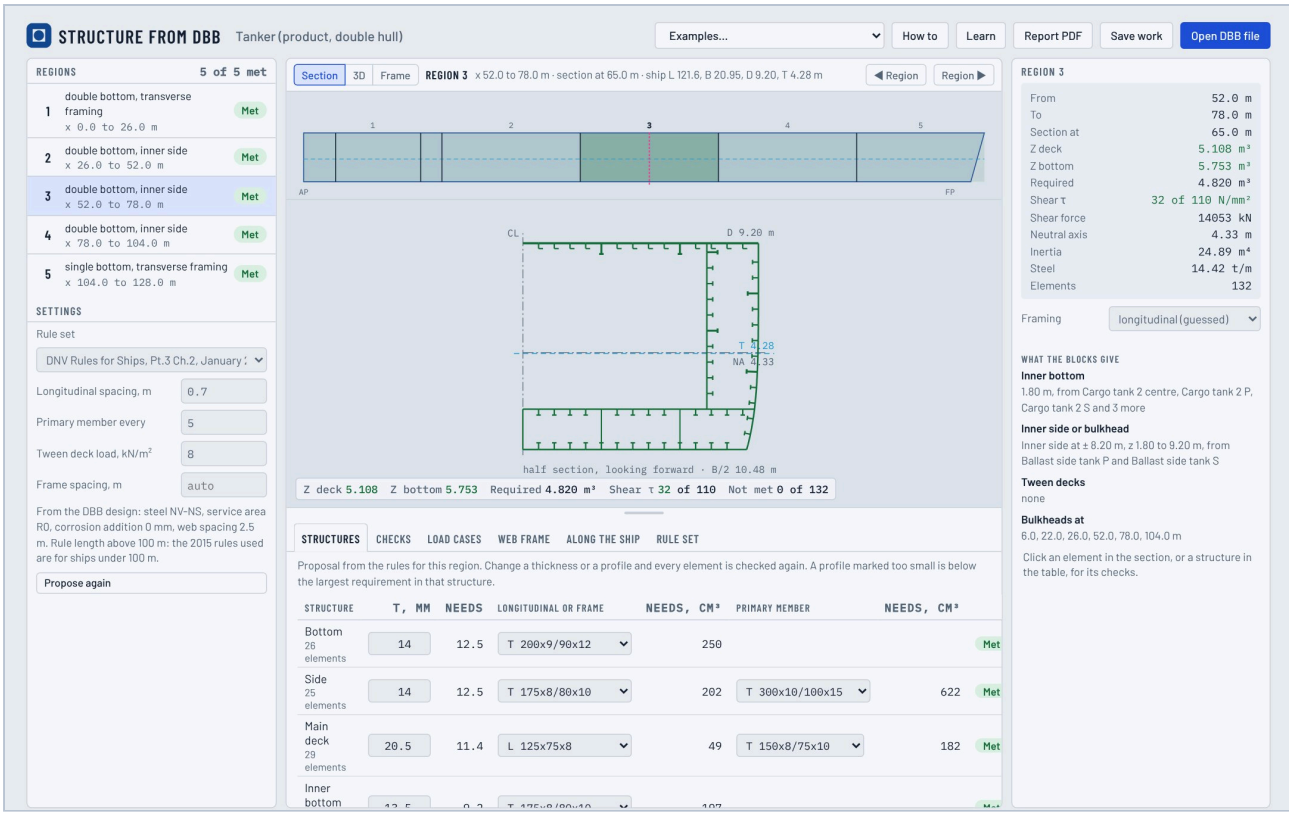


Figure 2. The double hull tanker, region 3: the cargo tank section with its double bottom and double side, the Structures table and the region readout.

Keys: left and right arrows step through the regions; Esc clears the selection. In 3D, drag to turn, right-drag to pan, scroll to zoom.

Settings from the design. Steel grade, service area, corrosion addition and web frame spacing come from the Structure tab of the DBB design. The left panel lists them; change them in the DBB editor.

Report and saving. *Report PDF* makes an A4 report of the whole ship: design and settings, the regions with their hull girder results, the section modulus along the ship, the bulkheads, the load cases, then one page per region with its section and web frame drawings, tables and the checks not met, and finally the method, assumptions, limitations and references. Every page carries the date, the app version and the design name. *Save work* keeps the DBB design together with your thicknesses, profiles, web frame sizes, settings and load cases in one .json file; *Open DBB file* opens it again. If the regions differ (for example after a change in the DBB editor), a new proposal is made and the load cases are kept.



Structure from DBB

Structural report: Tanker (product, double hull)

Proposal and checks of the longitudinal structure, the transverse web frames and the bulkheads of each region, from the Design Building Block arrangement. Rules: DNV Rules for Ships, Pt.3 Ch.2, January 2015 (L < 100 m). For educational use only: simplified estimates, not for the design, approval or construction of any ship (see Assumptions and limitations).

Design and settings

Item	Value	Item	Value
Rule length L	121.57 m	Steel grade, f1	NV-NS, 1.00
Breadth B	20.95 m	Service area	R0
Depth D	9.20 m	Corrosion addition	0.0 mm
Draught T	4.28 m	Web frame spacing	2.50 m
Block coefficient Cb	0.911	Longitudinal spacing	0.70 m
Speed	14.0 kn	Primary member every	5
Regions	5	Tween deck load	8.0 kN/m²

Rule length above 100 m: the 2015 rules used are for ships under 100 m, so the results are indicative only.

Regions

Region	x from, to (m)	Layout	Z deck (m²)	Z bottom	Required	Steel t/m	Frame usage	Status
1	0.0 to 26.0	double bottom 1.80 m, transverse framing	3.209	3.958	1.130	8.83	1.00	Met
2	26.0 to 52.0	double bottom 1.80 m, inner side, longitudinal framing	5.108	5.753	4.820	14.42	0.84	Met
3	52.0 to 78.0	double bottom 1.80 m, inner side, longitudinal framing	5.108	5.753	4.820	14.42	0.84	Met
4	78.0 to 104.0	double bottom 1.80 m, inner side, longitudinal framing	5.189	5.696	4.820	14.37	0.84	Met
5	104.0 to 128.0	single bottom, transverse framing	1.862	1.278	0.519	4.10	1.00	Met

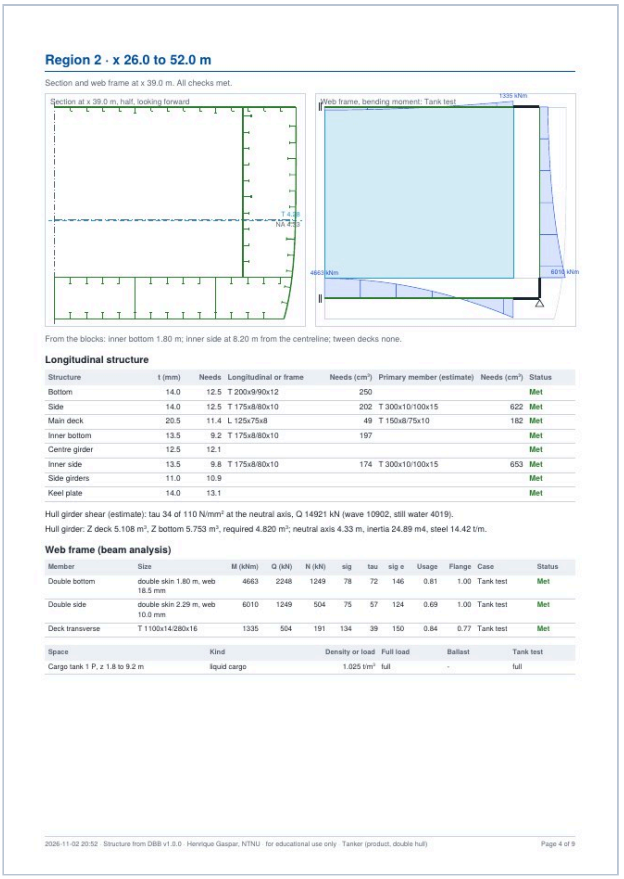
Hull girder section modulus along the ship

Bars: smaller of deck and bottom section modulus of each region. Orange: rule requirement along the length. Red: the largest requirement within each region.

Transverse bulkheads (simplified Pt.3 Ch.2 Sec.9, estimates)

x (m)	Type	Plating, bottom to top (mm)	Stiffeners	Needs (cm³)	Status
6.00	watertight	9 / 9 / 9 / 9 / 9	T 325x11/120x15	736	Met
22.00	watertight	9 / 9 / 9 / 9 / 9	T 325x11/120x15	736	Met
26.00	watertight	9 / 9 / 9 / 9 / 9	T 325x11/120x15	736	Met
52.00	watertight	9 / 9 / 9 / 9 / 9	T 325x11/120x15	736	Met
78.00	watertight	9 / 9 / 9 / 9 / 9	T 325x11/120x15	736	Met
104.00	collision	10 / 10 / 10 / 10 / 10	T 325x11/120x15	736	Met

First page: design, settings and the hull girder along the ship



Region 2 · x 26.0 to 52.0 m

Section and web frame at x 39.0 m. All checks met.

From the blocks: inner bottom 1.80 m; inner side at 8.20 m from the centreline; tween decks none.

Longitudinal structure

Structure	t (mm)	Needs	Longitudinal or frame	Needs (cm³)	Primary member (estimate)	Needs (cm³)	Status
Bottom	14.0	12.5	T 200x9/90x12	250			Met
Side	14.0	12.5	T 175x8/80x10	202	T 300x10/100x15	622	Met
Main deck	20.5	11.4	L 125x75x8	49	T 150x8/75x10	182	Met
Inner bottom	13.5	9.2	T 175x8/80x10	197			Met
Centre girder	12.5	12.1					Met
Inner side	13.5	9.8	T 175x8/80x10	174	T 300x10/100x15	653	Met
Side girders	11.0	10.9					Met
Keel plate	14.0	13.1					Met

Hull girder shear (estimate): tau 34 of 110 N/mm² at the neutral axis, Q 14921 kN (wave 10902, still water 4019).

Hull girder: Z deck 5.108 m³, Z bottom 5.753 m³, required 4.820 m³; neutral axis 4.33 m, inertia 24.89 m4, steel 14.42 t/m.

Web frame (beam analysis)

Member	Size	M (kNm)	Q (kN)	N (kN)	sig	tau	sig e	Usage	Flange	Case	Status
Double bottom	double skin 1.80 m, web 16.5 mm	4663	2248	1249	78	72	146	0.81	1.00	Tank test	Met
Double side	double skin 2.29 m, web 10.0 mm	6010	1249	504	75	57	124	0.69	1.00	Tank test	Met
Deck transverse	T 1100x14/280x16	1335	504	191	134	39	150	0.84	0.77	Tank test	Met

Space	Kind	Density or load	Full load	Ballast	Tank test
Cargo tank 1 P, z 1.8 to 9.2 m	liquid cargo	1.025 t/m³	full	-	full

A region page: section, web frame, tables and checks

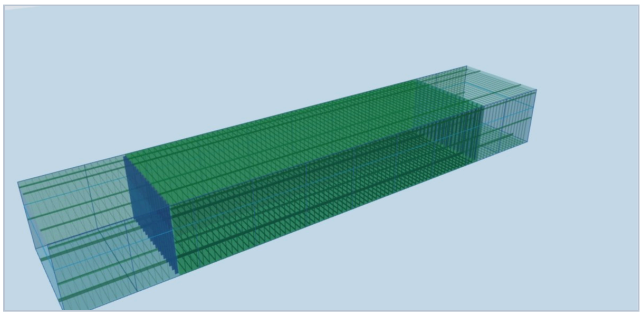
Figure 3. Report PDF for the tanker.

3. The examples

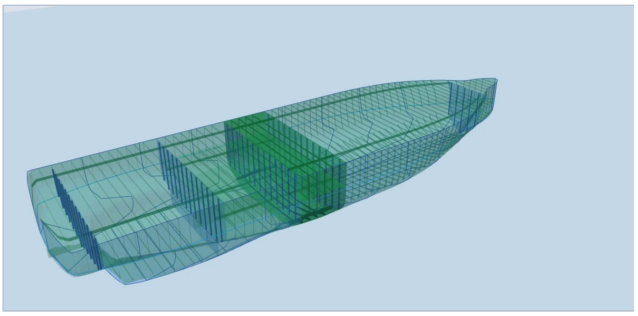
Seven designs come with the app. Each shows something different; the notes say what to look at and what the app leaves out for that ship.

Example	Size	What it shows	Watch for
Barge (Sollied 2009)	60 × 12 × 7.5 m, T 5 m	The hand calculation of the compendium: 1800 t lightship, 900 t ballast in each 10 m end tank, Q 600 t and M 9000 t m hogging. Three regions, transverse framing, single bottom. The lessons use it.	The full-breadth end tanks in the tank test give deep floors (about 1.4 m); a centreline girder would halve their span.
R/V Gunnerus	36 × 9.6 × 4.3 m, T 2.5 m	A real research vessel hull from the open Gunnerus data, 45 blocks. Six regions, all transversely framed (L < 60 m); a double bottom only under the engine room; tween decks.	The hull girder demand is small, so minimum thicknesses govern. The hidden buoyancy hull is a cleaned copy of the sliced 3D model.
PSV	84 × 16.4 × 7.9 m, T 4.9 m	Brine and drill water tanks as liquid cargo, dry bulk, engine room; longitudinal framing amidships, transverse at the ends; 1.5 m double bottom.	Sagging governs. The tank test of the full-breadth cargo tanks drives the double bottom.
AHTS	85 × 21.3 × 9.5 m, T 5.0 m	A longitudinal bulkhead in region 2, a tween deck at 7.5 m, chain lockers and machinery.	Wide and short: the deck transverses and side webs carry a lot; compare the frame of region 2 with region 4.
Double hull tanker	122 × 21 × 9.2 m, T 4.3 m	Cargo tanks inside a double side (inner side at 8.2 m) and a 1.8 m double bottom; the double side is a double skin beam in the frame.	Above 100 m the 2015 rules are indicative only. The deck reaches about 21 mm.
Feeder container ship	131 × 22.9 × 10.2 m, T 5.0 m	Container holds with a double bottom and a deep hull girder demand.	Hatch openings are not modelled, so the deck section modulus is too high: the real deck needs more steel at the hatch coamings.
Handysize bulk carrier	145 × 25 × 12.5 m, T 6.8 m	Five holds, deck up to about 30 mm, shear stress close to the limit in region 2.	No hopper or topside tanks and no hatch openings; read it as a box-shaped hold.

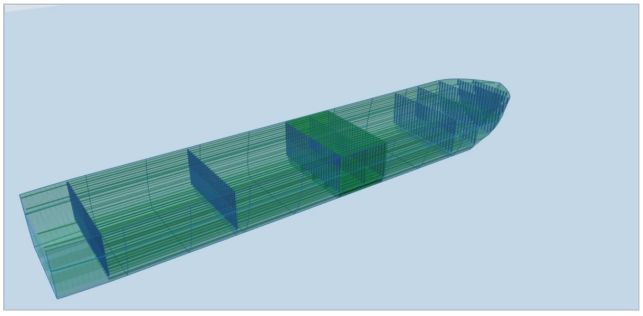
The end regions reach the ends of the hull. They are judged at their own section, where the rule moments are small; forward of the forward perpendicular the rule moments, and so the required section modulus, are zero.



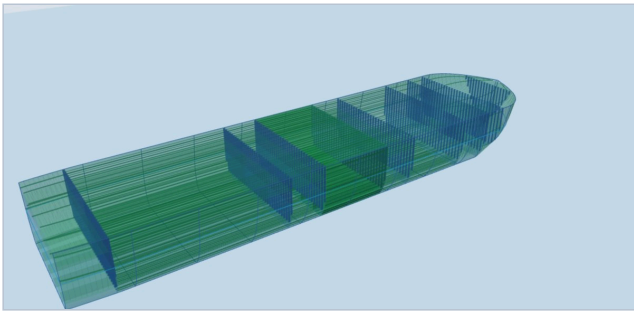
Barge (Sollied 2009)



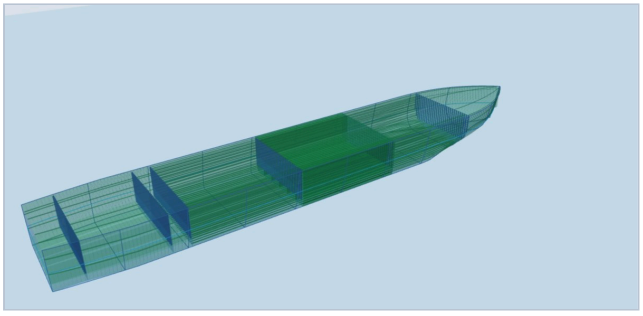
R/V Gunnerus



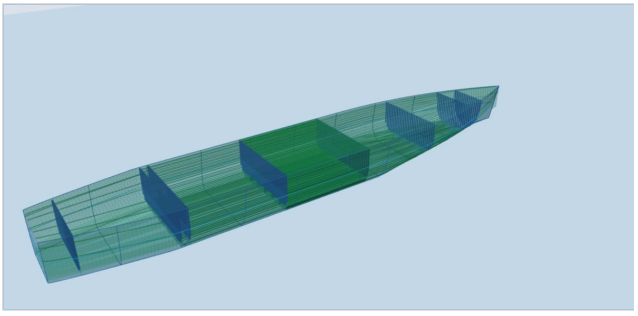
PSV



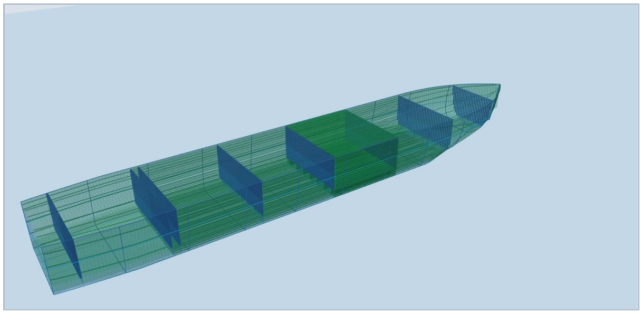
AHTS



Double hull tanker



Feeder container ship



Handysize bulk carrier

Figure 4. The seven examples in the 3D view, with the structure of every region; the selected region is in full colour.

4. How the app reads a DBB design

The arrangement sets the structure: the transverse bulkheads cut the ship into regions, and the blocks at each region's middle say where the inner bottom, inner side and tween decks are.

Coordinates. As in the DBB editor: metres, x forward from the aft end, y to port, z up from the baseline. Main dimensions (L, B, D, T, Cb), the rule length and the speed come from the design through the DBB core.

Blocks used. Blocks inside the hull and below the main deck. A block named *Hull*, a block larger than 40% of the hull volume, and hidden blocks are ignored.

Transverse bulkheads. A bulkhead is placed where the ends of tall blocks (taller than 30% of D) meet across at least 60% of the breadth. Block ends within 1.5 m are taken as one bulkhead, and ends within 2% of L from either end of the ship are ignored.

Regions. The ship is cut at the bulkheads, from the aft end to the fore end of the hull at the deck (the overhangs aft of AP and forward of FP belong to the end regions). A region shorter than 5% of L is merged with its neighbour. Each region gets one section at its middle, on the real hull shape.

Inner boundaries from the blocks, at the region's middle:

- **Inner bottom:** the top of a layer of blocks at the bottom (tanks from the baseline, or blocks that start at one height) covering at least half the breadth. The right panel lists the blocks used.
- **Inner side or longitudinal bulkhead:** the side faces of tall blocks running along the region. Near the shell it is an inner side; further in, a longitudinal bulkhead.
- **Tween decks:** heights where block tops or bottoms line up across the breadth, between the inner bottom and the main deck.

Shell zones. The half section of the shell is split by its slope:

- the **bottom** runs from the keel (the lowest height with breadth) until the shell is steeper than 30° for two steps;
- the **side** starts where it is steeper than 70° and stays steeper than 55° up to the deck;
- the **bilge** lies in between.

The app reads the design once on opening. *Propose again* repeats the proposal with the current settings; to change the arrangement, edit it in the DBB editor and open it again.

5. Longitudinal structure

Each region's section is filled with plates and longitudinals, sized to DNV Rules for Ships Pt.3 Ch.2 (January 2015, ships under 100 m), and the hull girder section modulus is checked against the rule value.

Structures and elements. The structures of a section are the bottom, bilge, side, main deck, inner bottom, centre girder, side girders, inner side or longitudinal bulkhead, and tween decks. Each structure is divided into:

- **plates** between longitudinals, with the thickness of their structure; the first bottom plate is the **keel strake**;
- **longitudinals** at the longitudinal spacing s (0.7 m by default), all of one profile per structure;
- a **primary member** at every n-th longitudinal (every 5th by default; none on the bilge). In a double bottom it is a **side girder** plate from the bottom to the inner bottom.

Profiles. Flat bars, unequal angles (EN 10056 sizes, 100×65×7 to 250×90×12) and built T bars, drawn with their real web and flange. The section modulus of a profile includes its attached plate.

Local pressures (kN/m²):

Structure	Pressure
Bottom, bilge, side	Sea: below the waterline p1 = 10 h0 + pdp, above it p2 = pdp − (4 + 0.2 ks) h0, at least 6.25 + 0.025 L (Sec.4)
Main deck	Weather deck: 0.8 (pdp − 4.4 (D − T)), at least 5
Inner bottom, inner side, longitudinal bulkhead, girders	Tank head to the deck plus 0.76 m: 10 · 1.025 · (D + 0.76 − z)
Tween decks	The tween deck load in Settings (8 kN/m² by default)

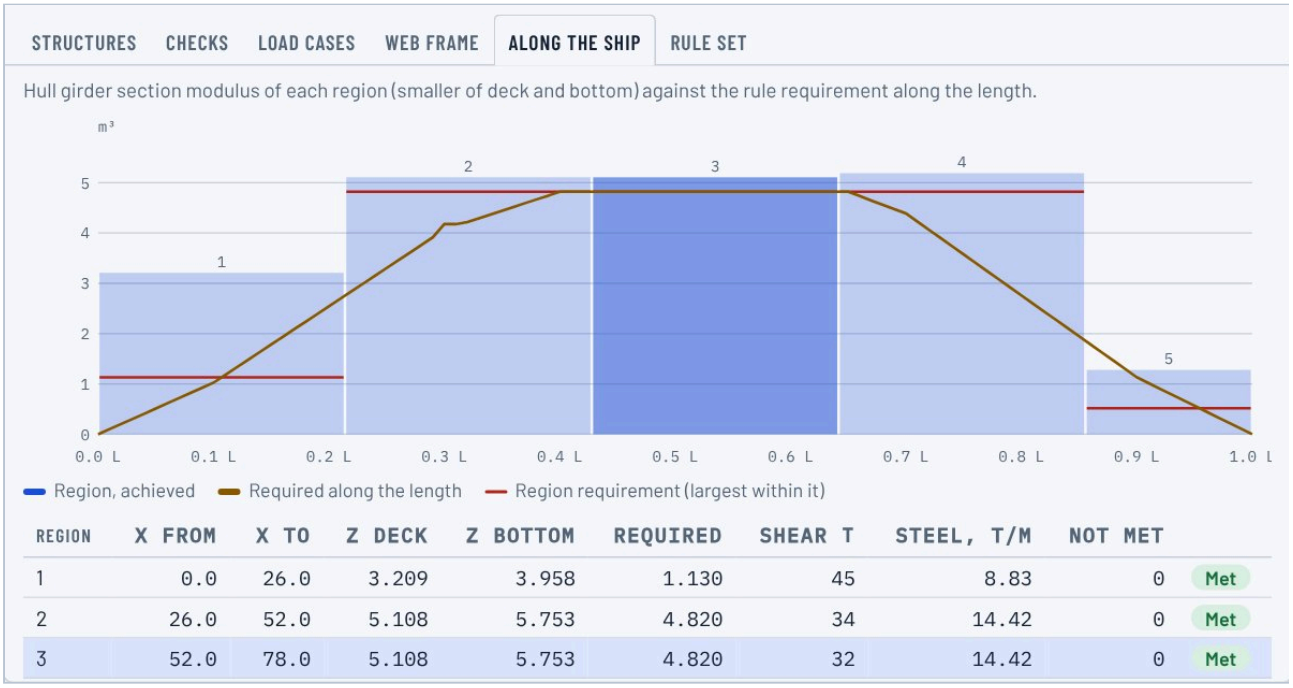
Rules applied, one entry each in the Rule set tab:

Rule	Requirement	Reference	Basis
Plate, lateral pressure	t = 15.8 ka s √(p / (σ f1)) + tk, σ = 120 (160 for tween decks and bulkheads)	Sec.6 C, 7 C, 8 C	rule
Plate, minimum	t = t0 + k L / √f1 + tk, per structure	Sec.6 C, 7 C	rule
Keel plate	t = 7.0 + 0.05 L + tk	Sec.6 B	rule
Longitudinal, section modulus	Z = 83 l² s p / (σ f1), at least 15 cm³	Sec.6 C, 7 C, 8 C	rule
Longitudinal, web thickness	tw ≥ 4.5 + 0.01 L (or 1.5 + hw/g) + tk	Sec.5 C600, 6 C300	rule
Primary member, section modulus	Z = 100 S² b p / (160 f1)	general girder formula	estimate
Hull girder section modulus	Z at deck and bottom ≥ Zreq at that x	Sec.4 C	rule

Here l is the web frame spacing (the span of a longitudinal), b the breadth carried by a primary member and S its span.

Stress in longitudinals. The allowed stress in a longitudinal falls from 160 near the neutral axis to 95 at the deck and the bottom when the section modulus just meets the rule. It rises back towards 160 as the section modulus exceeds the rule, so a stronger hull girder allows lighter longitudinals.

Hull girder. The required section modulus follows the rule still-water and wave bending moments along the length. Within 0.3 to 0.7 L it is at least the minimum Z0. A region takes the largest requirement within it; the two end regions, whose section lies in the fine ends of the hull, are judged at their section instead. The app sums plates and profiles to find the neutral axis, the inertia, and Z at the deck and the bottom. The *Along the ship* tab plots each region against the rule curve.



REGION	X FROM	X TO	Z DECK	Z BOTTOM	REQUIRED	SHEAR T	STEEL, T/M	NOT MET	
1	0.0	26.0	3.209	3.958	1.130	45	8.83	0	Met
2	26.0	52.0	5.108	5.753	4.820	34	14.42	0	Met
3	52.0	78.0	5.108	5.753	4.820	32	14.42	0	Met

Figure 6. Along the ship: the section modulus of each region against the rule requirement along the length.

The proposal. For each structure the app takes the thickest plate any element needs and the lightest profile that meets the largest section modulus, then raises the deck and the bottom in 0.5 mm steps until the hull girder is met. It repeats this a few times, because the allowed stress depends on the section.

Editing. In the *Structures* tab, change a thickness or pick a profile; profiles below the requirement are marked *too small*. Every element is checked again at once. The *Checks* tab lists what is not met; click an element in the section for its log.

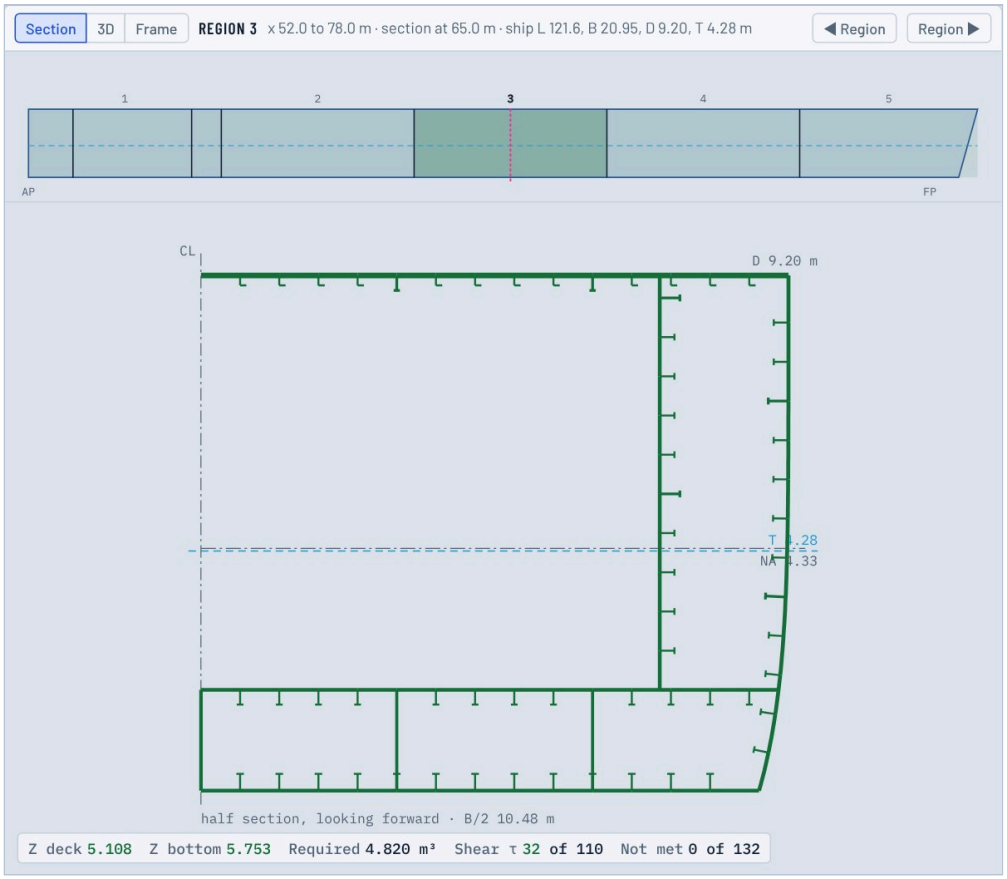


Figure 5. Half section of the tanker cargo region: plates, longitudinals, side girders and the double side; the readout gives Z at deck and bottom against the requirement and the shear stress.

Transverse framing, buckling and hull girder shear

Each region is framed longitudinally or transversely, and buckling and hull girder shear are checked. All three are simplified and marked as estimates.

Framing. The app guesses the framing per region:

- **transverse** for ships under 60 m, and for regions whose middle is within 0.15 L of either end;
- **longitudinal** elsewhere.

Change it for a region in the right panel. That region then gets a new proposal, and the others keep their edits. The frame spacing is in Settings; it is 0.48 + 0.002 L by default.

In a transversely framed region:

- the plating spans between frames, so its short side is the frame spacing;
- there are no ordinary longitudinals, but the girders and stringers (every n-th position) stay;
- floors (single bottom), side frames, deck and tween beams and bulkhead frames are sized with the general stiffener formula of the rules. The span runs between decks, or between girders when these are closer.

$$Z = \frac{100\, l^2\, s\, p}{160\, f_1}$$

The end brackets of each frame are estimated as a = 75 √(Z / t), with t the larger of 6.5 mm and the frame web. The frames are dashed in the section and drawn at every frame spacing in 3D. The Structures table shows the frame profile in the *Longitudinal or frame* column.

Buckling. The check is simplified, after the classic DNV Pt.3 Ch.1 Sec.13. The compressive stress σa comes from the design hull girder moments: sagging above the neutral axis, hogging below.

- **Plating:** σel = 0.9 k E (t / s)², with k = 4 for longitudinal framing or 1.21 (1 + (s/l)²)² for transverse framing. It is written as a required thickness.
- **Longitudinals as columns:** σel = 0.001 E Ia / (A I²), with a plate flange of 0.8 s.
- **Johnson correction:** above σf / 2 it gives σc = σf (1 − σf / (4 σel)), with σf = 235 f1.
- **Requirements:** σc ≥ σa for plates and σc ≥ σa / 0.85 for longitudinals.

The proposal meets both. This is why transversely framed plating near midship comes out thicker.

Hull girder shear. The shear stress is taken at the neutral axis:

$$\tau = \frac{Q\,S}{I\,\Sigma t} \le 110\,f_1$$

Here S is the first moment of the area above the neutral axis, and Σt the thickness of the vertical plating cut there (sides, inner sides, bulkheads).

- **Q** is the wave shear in the IACS UR S11 form, 0.3 F C L B (Cb + 0.7), plus the still-water shear from the weights and buoyancy of the DBB design. If the design does not float yet, the still-water shear is taken as 5 Ms / L.
- When the stress is too high, the proposal thickens the side plating.
- The readout under the section and the right panel show τ against the allowed value. *Along the ship* lists τ per region.

Plating as flange, lightening holes and the moment along the length

Three items come from the compendium *Strukturdesign av skip* (Sollied, 2009, Høgskolen i Ålesund). All are marked as estimates.

- **Plating as flange of the web frames.** The deck, bottom, inner bottom, side and inner side plating are the flanges of the web frame girders. The frame bends the plating across the ship while the hull girder compresses it along the ship. The stress across is taken from the frame forces, σay = |N| / A + |M| d / I, with d the distance from the girder's neutral axis to the plating. The stress along, σax, is the design hull girder stress in sea cases and only its still-water share in the other cases (tank test, harbour). The two are combined with the DNV biaxial interaction formula:

$$\frac{\sigma_{ax}}{\sigma_{cx}} - K\frac{\sigma_{ax}\,\sigma_{ay}}{\sigma_{cx}\,\sigma_{cy}} + \left(\frac{\sigma_{ay}}{\sigma_{cy}}\right)^{n} \le 1, \qquad K = c\,\beta^{a}, \qquad \beta = \frac{1000\,s}{t}\sqrt{\frac{\sigma_y}{E}}$$

- σcx and σcy are the critical stresses of the panel along and across, with the Johnson correction. For l/s < 1.5: c = 0.78, a = −0.12, n = 1.0; otherwise c = 0.80, a = 0.04, n = 1.2.
- **What the proposal does.** The girders are sized for their stresses only: a deeper girder is the wrong cure for buckling plating. Where a flange still fails the check, the proposal thickens that plating in steps of 0.5 mm, then sizes the frame again. In the examples this mainly hits double bottoms under the tank test of full-breadth tanks. A real design would often add buckling stiffeners or a centreline girder instead; the student can try that by hand.
- **Lightening holes.** In the Web frame tab each member has a *Hole Ø, mm* field. A round hole of diameter d in a web of height h keeps r = 1 − d / h of the shear area. The right panel shows r, and the work file keeps the holes.

- **Still-water moment along the length.** It follows the rule distribution: 0 at the ends, 0.15 at 0.1 L and 0.9 L, and full from 0.3 L to 0.7 L. The wave moment is full from 0.4 L to 0.65 L.
- **The barge example.** *Barge, hand calculation (Sollied 2009)* is a 60 × 12 × 7.5 m box at 5 m draught, with 1800 t of lightship spread evenly and 900 t of ballast in each 10 m end tank. The app gives the hand values: shear force 600 t (5884 kN) and bending moment 9000 t m (88 260 kNm). It is a good first exercise: do it by hand, then compare.

6. Web frame analysis

Each region's transverse web frame is analysed as a plane frame of beams, the method of the 3D Beam program in Nauticus Hull, and its girders are sized from the computed forces.

The model. One web frame at the region's middle, port half, built from the section:

Member	Idealised as	Plate flange
Double bottom	Double skin beam at half the double bottom height; web = the floor	Bottom and inner bottom plating
Double side (inner side from inner bottom to deck)	Double skin beam on the mid line of the double side	Side and inner side plating
Side web frame (single side)	Built T girder along the real shell, from the inner bottom to the deck	Side plating
Floor (single bottom)	Built T girder along the bottom shell	Bottom plating
Deck transverse, tween beams	Built T girder	Deck plating
Longitudinal bulkhead web	Vertical built T girder, when the bulkhead meets the double bottom, the deck or a tween deck	Bulkhead plating

- **Rigid ends** at the knuckles: half the double bottom height at the foot of the side, and half the double side width at the ends of the double bottom and deck. They are drawn dark.
- **Supports:** symmetry at the centreline (no sideways move, no rotation), and a vertical support where the side meets the bottom and at the foot of a bulkhead web on the double bottom. These stand for the shear the side shell and bulkheads carry to the transverse bulkheads.
- **Effective flange** of the plating: be = min(web frame spacing, 0.3 × member length).

Section properties. Area A, inertia I and the smaller section modulus W are taken from the plate flange, the web and the face plate. The shear area follows 3D Beam:

$$A_z = \frac{I\, t_w}{S_y}$$

where tw is the web thickness and Sy the first moment of area above the neutral axis.

Solver. The frame uses Timoshenko beam elements (shear deformation included) and the matrix displacement method, R = K r. Distributed loads vary linearly along each element and become equivalent nodal loads. Forces are recovered at seven points per element. For every load case, the sum of the reactions equals the applied load; the Web frame tab shows both.

Results in the Frame view. Pick a load case, tick *Spaces* and *Loads* to see how it loads the frame (section 7), and pick a diagram: bending moment, shear force, axial force or deformed shape. The largest value of each member is labelled. Members are drawn as bands as deep as their web and coloured by their check. Click a member for its properties and checks.

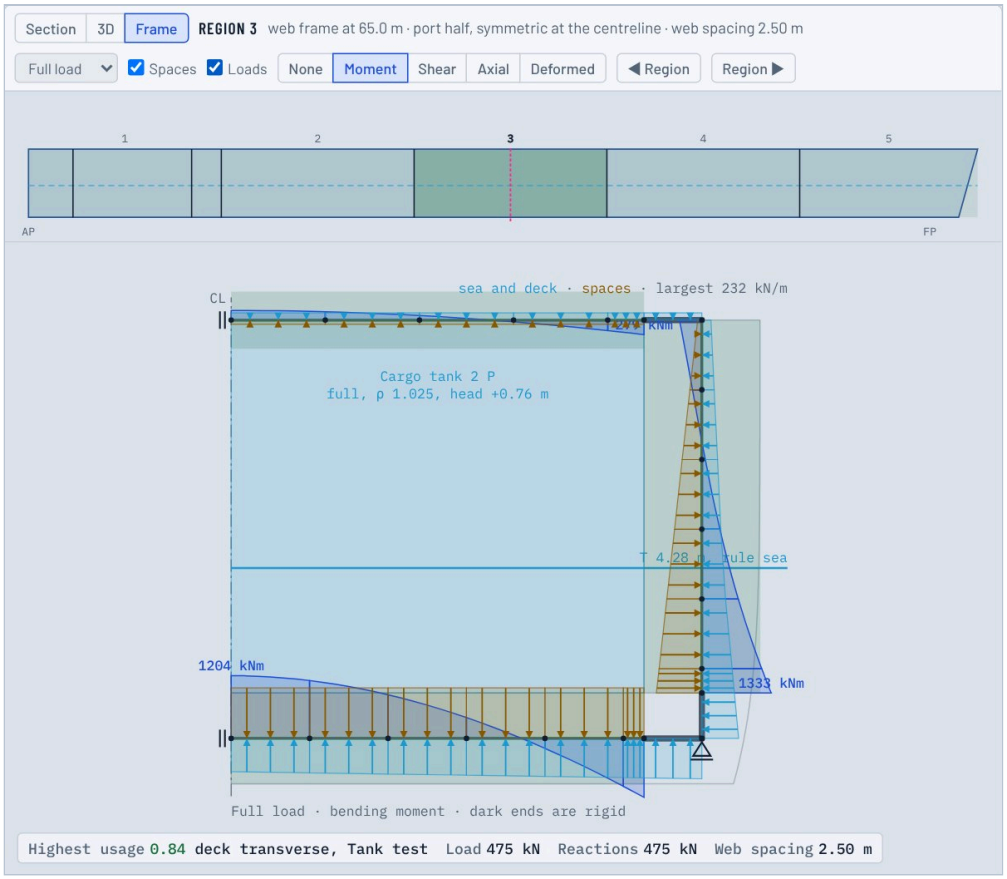


Figure 7. The web frame of the tanker cargo region as a beam model: the full load case with Spaces and Loads ticked and the bending moment diagram.

Checks, against the ticked load cases:

Check	Formula	Allowed
Normal stress	σ = |N| / A + |M| / W, with the smaller W	160 f1
Shear stress	τ = |Q| / Az	90 f1
Equivalent stress	σe = √(σ² + 3 τ²)	180 f1 (assumed)
Web thickness	tw ≥ 6 + 0.04 L / √f1 + tk for floors, 5 + 0.02 L / √f1 + tk for other webs	assumed

Usage is the largest ratio of stress to allowed stress over all members and cases. A region is met only when its elements, its hull girder and its web frame are all met.

Sizing. The proposal picks the lightest built T girder (webs 200 to 2000 mm) or double skin web thickness (6 to 30 mm) that carries the forces. It solves the frame again, because stiffer members attract more moment, until the sizes repeat, then steps up any member that still fails. In the *Web frame* tab you can pick another size for any member (sizes that would fail are marked *too small*) and the frame is solved again. *Propose this frame* sizes it again.

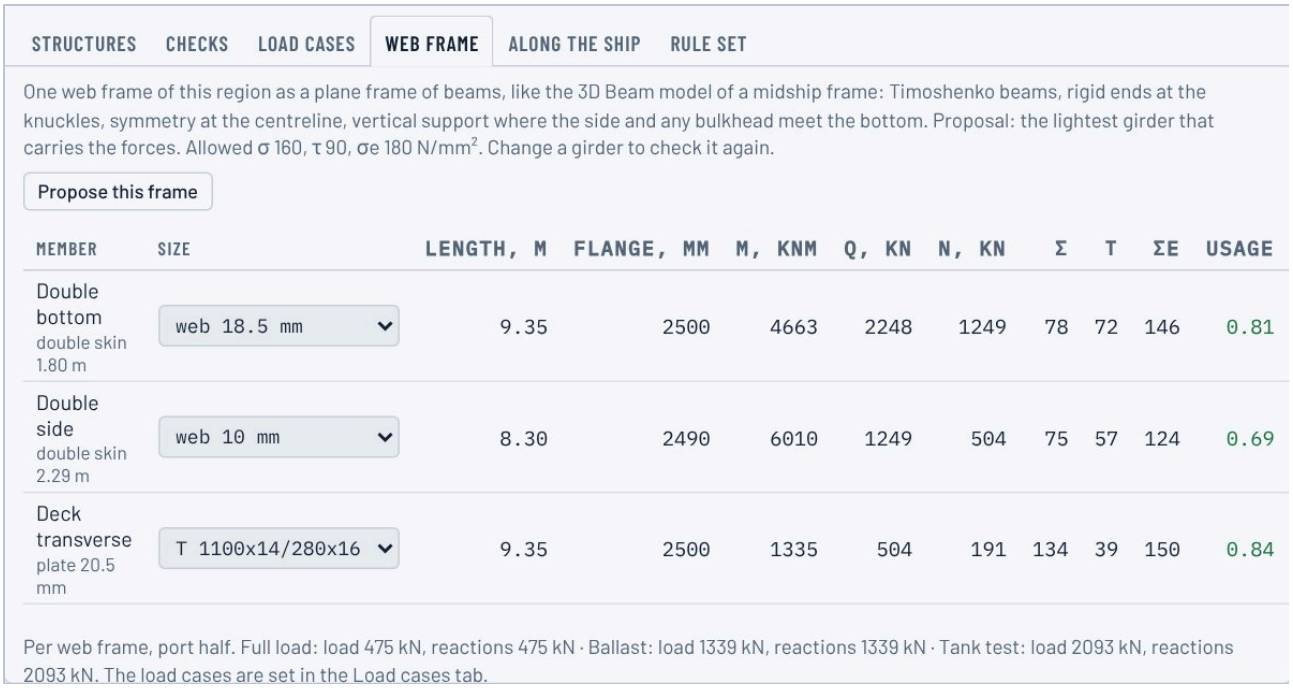


Figure 8. The Web frame tab: each member with its size, forces, stresses, usage, plating as flange and lightening holes.

7. Load cases

The web frames are loaded by load cases that start from three scenarios read from the blocks (full load, ballast and tank test) and that you can edit, extend or switch off in the *Load cases* tab.

Spaces. The frame is split into spaces:

- from the inner bottom (or the bottom) to the deck, divided at each tween deck;
- from the centreline to the side, or to the inner side when there is a double side;
- divided again at a longitudinal bulkhead that joins the frame.

Each space takes the block that covers at least a quarter of it at the region's middle. Spaces inside the double bottom and the double side are not listed: their pressure acts on both skins of the same beam and balances inside it.

What fills a space. Read from the block's group, name and requirements:

Kind	How it is recognised	Load on the frame
Liquid cargo	Cargo group or a liquid cargo requirement, with *tank* in the name (not *dry* or *bulk*)	Liquid head, the block's density (1.025 t/m³ if none)
Dry cargo	Cargo group or requirement, otherwise	The block's DBB mass on its floor
Ballast	*ballast*, *WB*, *peak*, or any other tank	Liquid head, 1.025 t/m³
Fuel oil	*fuel*, *FO*, *MGO*, *HFO*, *lub*, *diesel*	Liquid head, 0.85 t/m³
Fresh water	*water*, *FW*, *DW*	Liquid head, 1.0 t/m³
Dry space	Any other block (machinery, stores, accommodation)	The block's DBB mass on its floor
Void	*void*, *cofferdam*	Nothing
No block	No block covers the space	The tween deck load in Settings

A liquid gives a pressure that grows with depth below the tank top plus the head above it:

$$p\left(z\right) = 10\,\rho\,\left(z_{top} + h_{above} - z\right)$$

It acts on every wall of the space: down on its floor, up on its ceiling, outwards on the side or double side, and on either face of a bulkhead. A dry space puts its DBB mass on its floor:

$$q = \rho_{block} \cdot \frac{fill}{100} \cdot g \cdot h_{block}$$

The sea acts on the shell at the case draught: the rule pressure with waves (p1, p2 of Sec.4, with T set to the case draught) or still water, 10 · 1.025 · (T − z). The green sea on the weather deck uses the rule deck load at that draught. Every pressure is multiplied by the web frame spacing to give a line load in kN/m.

The three scenarios

	Full load	Ballast	Tank test
Draught	Design T	About 2 + 0.02 L, between 0.5 T and 0.8 T	No sea

	Full load	Ballast	Tank test
Sea	Rule, with waves	Rule, with waves	None
Green sea on deck	Yes	Yes	No
Head above tank tops	0.76 m	0.76 m	2.4 m
Ballast tanks	Empty	Full	Full
Liquid cargo	Full	Empty	Full
Fuel and fresh water	Full	Full	Full
Dry cargo	Loaded	Empty	Empty
Dry spaces, no block	Loaded	Loaded	Empty

Editing. The *Load cases* tab has one column per case, shared by every region:

- the case name; *Checked* (used in the checks and the proposal); draught; sea (rule, still water, none); green sea; head above the tank tops; a factor on the allowed stresses;
- one row per space of the selected region, with its density (t/m³) or deck load (kN/m²), and per case whether it is full or loaded.

Add case copies the case shown in the Frame view, × removes a case, and *Reset to the scenarios* starts again from the three scenarios. A change keeps the girders and checks them again, so you see its effect. Press *Propose web frames again* (all regions) or *Propose this frame* to size them for the new cases.

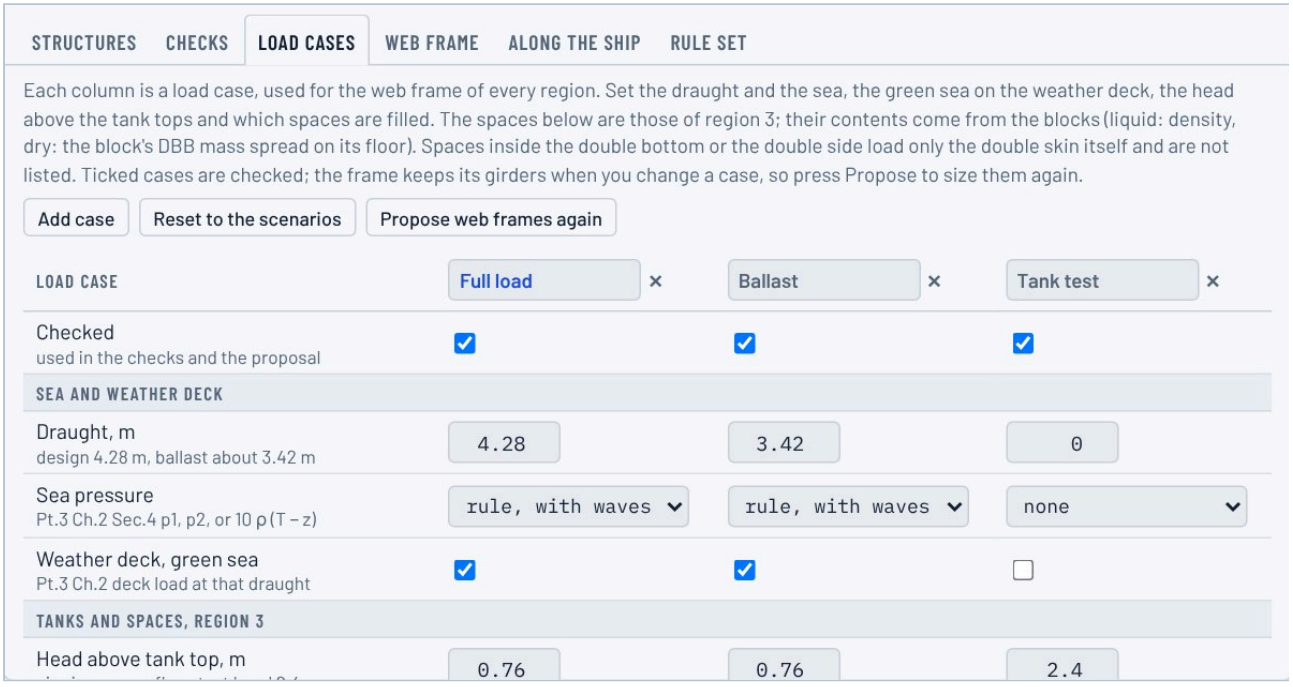


Figure 9. The Load cases tab: one column per case, the sea and weather deck, then the tanks and spaces of the region.

Seeing a case. In the Frame view, choose the case and tick:

- **Spaces:** each space shaded as the case fills it (blue liquid, amber dry, dashed empty), with its block, density or load, and the case draught;

- **Loads:** the loads on each beam as trapezoids with arrows. Blue comes from outside (sea, green sea) and amber from inside (the spaces), so you see how a full tank at sea offsets the sea pressure.

The readout under the drawing gives the net load and the reactions of the case.

8. Transverse bulkheads and the 3D view

Bulkheads. Every transverse bulkhead found in the blocks (section 4) is sized with a simplified version of Pt.3 Ch.2 Sec.9. The results are listed under *Along the ship* and marked as estimates.

- Watertight, loaded by a head of water to the deck: p = 10 h.
- **Plating** in strakes of about 2 m: t = 15.8 ka s √(p / (σ f1)) + tk with σ = 220, and at least 5 + 0.03 L / √f1 + tk.
- The foremost bulkhead beyond 0.8 L is taken as the **collision bulkhead**: σ = 160 and at least 6 + 0.03 L / √f1 + tk.
- **Vertical stiffeners** at the longitudinal spacing, spanning from the inner bottom (or the keel) to the deck: Z = 83 l^2 s p / (160 f1), at least 15 cm^3, with p at mid span. The lightest profile that meets it is chosen.

3D view. The *3D* button shows the hull on the DBB editor's own engine, with the plates, longitudinals and primary members of every region and the bulkheads.

- Within a region the elements keep the share of the girth they have at the region's section, so they follow the hull station by station towards the ends.
- Profiles are drawn in their real shape: flat bar, angle or T, with web and flange.
- The selected region is in full colour and the others are faded. Clicking a region in 3D selects it.
- *Hull*, *Structure* and *Bulkheads* show or hide each part; the camera buttons give 3D, top, side, front, fit and region views.

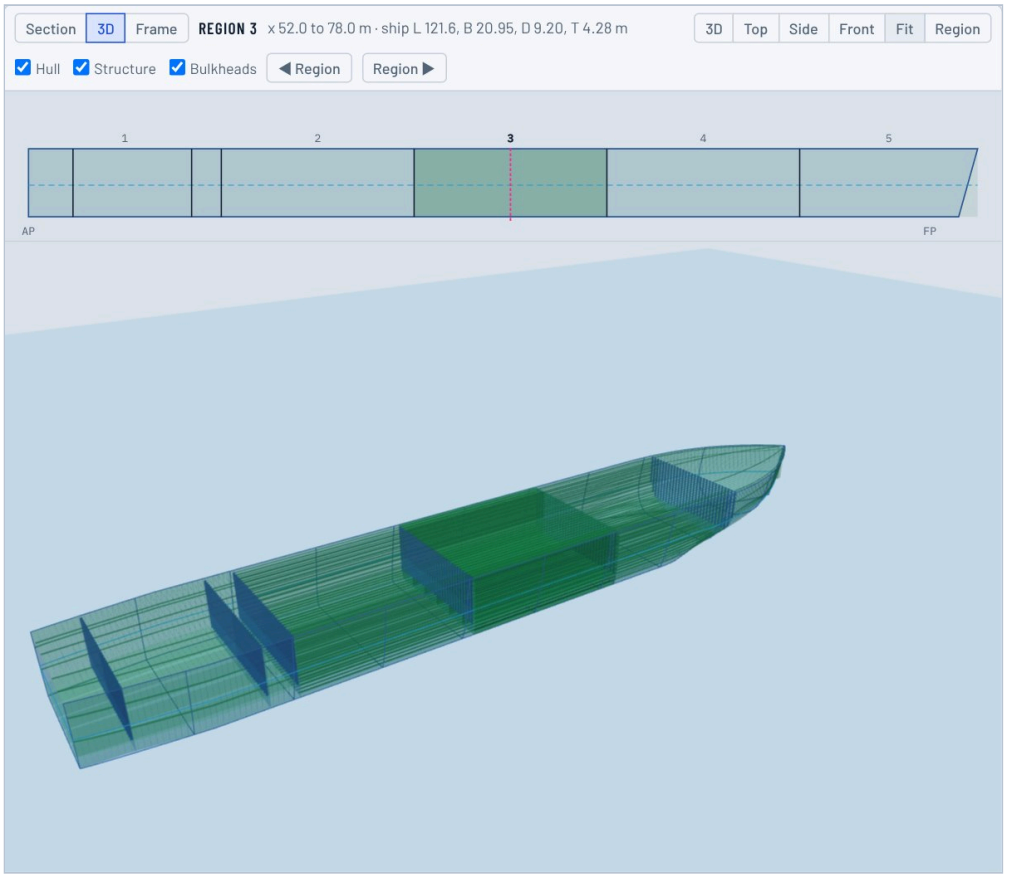


Figure 10. The 3D view of the tanker: plates, longitudinals, primary members and bulkheads of every region on the real hull.

Transverse frames are drawn in 3D at every frame spacing; the web frames of the beam model are shown in the Frame view only. A double side or inner side is drawn at its own breadth.

9. Assumptions

The app fills the gaps between a block arrangement and a structural design with the assumptions below. Each is a choice a student can question, and many can be changed in the app.

Topic	Assumption	Can you change it?
Rules	DNV Rules for Ships Pt.3 Ch.2, January 2015, for ships under 100 m; used above 100 m with a warning	No (more rule sets can be added as entries)
Framing	Guessed per region (transverse for ships under 60 m and near the ends, longitudinal elsewhere); can be changed per region in the right panel	No
Material	Grade, service area and corrosion addition from the DBB design; f1 from the grade	In the DBB editor
Regions	One section per region, at its middle; the structure is constant within a region	No
Arrangement	Bulkheads, inner bottom, inner side and tween decks are where the blocks say	In the DBB editor
Spacing	Longitudinals at 0.7 m; a primary member at every 5th; web frames at the spacing in the design	Settings, DBB editor
Tank pressure (plating)	Head to the deck + 0.76 m, no accelerations of the contents	No
Primary members	General girder formula, span = web frame spacing	No; marked as estimate
Allowed stress in longitudinals	160 near the neutral axis to 95 at deck and bottom when Z = Zreq	No
Web frame model	Port half, symmetric; Timoshenko beams; rigid ends at the knuckles	No
Supports	Symmetry at the centreline; vertical support where the side and a joined bulkhead meet the bottom	No
Effective flange	min(web frame spacing, 0.3 × member length)	No
Web frame allowed stresses	σ 160 f1 and τ 90 f1 (the girder values of Pt.3 Ch.2); σe 180 f1 assumed	Per case, a factor
Minimum web thickness	Floors 6 + 0.04 L / √f1 + tk, other webs 5 + 0.02 L / √f1 + tk (assumed)	No

Topic	Assumption	Can you change it?
Normal stress	|N|/A + |M|/W with the smaller W, on the safe side	No
Space contents	Recognised from block group, name and requirements; liquid densities by kind; dry spaces carry their DBB mass	Load cases tab
Liquid head	To the space top + 0.76 m at sea, + 2.4 m in the tank test	Load cases tab
Ballast draught	About 2 + 0.02 L, between 0.5 T and 0.8 T	Load cases tab
Tank test	Judged at the same allowed stresses as at sea (factor 1.0)	Load cases tab
Bulkheads	Watertight, head to the deck; foremost one beyond 0.8 L is the collision bulkhead	No; marked as estimate
End regions	Reach the ends of the hull; judged at their own section, with the rule moments at that position (zero forward of FP)	No
Plating as frame flange	Girders are sized for stresses; a flange that buckles is cured by thicker plating, not a deeper girder	By hand, in the Structures and Web frame tabs
Profiles	Flat bars, EN 10056 angles and built T bars; no bulb profiles	No

10. Limitations and intended use

Structure from DBB is for educational use only: teaching and early concept exploration. Its results are simplified estimates and must not be used for design approval, class or flag-state documentation, or any safety-critical decision.

What it does not do:

- **Buckling and shear only in simplified form; no fatigue, vibration or ultimate strength.** Buckling is the classic elastic check with the Johnson correction, without lateral-torsional buckling, and the hull girder shear stress is taken at the neutral axis only.
- **No finite element analysis.** The web frame is a beam idealisation of one frame. The effective flanges, rigid ends and supports are assumptions that can move the stresses by about 20%.
- **Transverse framing in simplified form (general stiffener formula, estimated brackets),** no stringers, no deck girders, no pillars, no hatch openings, no welds. Cut-outs only as round lightening holes in the web frames (shear area only, no stress concentration).
- **One section and one web frame per region.** Changes of shape within a region (towards the ends, the bilge running out) are not followed by the scantlings.
- **Symmetric loads only.** The half model cannot take one wing tank full and the other empty, or a heeled ship.
- **No dynamic tank pressures.** The liquid heads have no accelerations from ship motions, and no sloshing.
- **Limited hull girder checks.** Section modulus and a simplified shear stress: no torsion or hull girder ultimate strength. The still-water moment is the rule minimum unless the DBB design provides one.
- **Bulkheads are simplified.** Plating and vertical stiffeners only, without girders or the full Sec.9 cases.
- **The rule edition.** The 2015 rules for ships under 100 m are used; above 100 m the results are indicative only.
- **The reading of the blocks.** It depends on how the arrangement is modelled. Tank blocks named or grouped unusually may be read as the wrong kind; check the Load cases tab and correct the kind by its density or load. Full-breadth tanks with no centreline bulkhead give deep girders, as the model has nothing else to carry them.

Known open questions:

- The rule allowance for tank test conditions is not confirmed, so the tank test uses a factor of 1.0.
- The equivalent stress limit 180 f1 and the minimum web thickness for frames are assumed values.
- The DBB core used here has one fix that the DBB editor still needs: the still-water moment had the wrong sign, so hogging was taken as sagging.

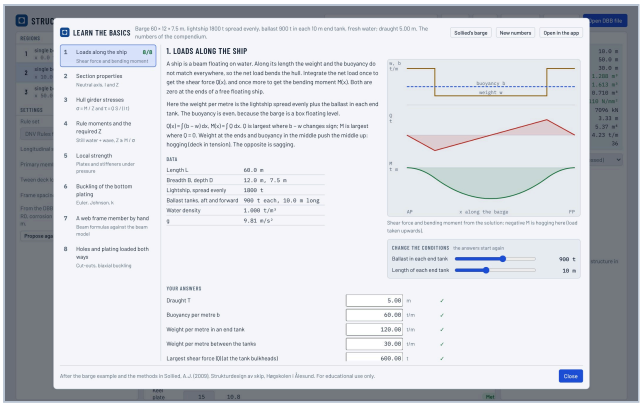
11. Learn the basics

The **Learn** button, beside How to, opens eight short lessons on a box barge. They follow the barge example and the methods in Sollied (2009). Each lesson gives the principle and the data. The student works the numbers by hand, types the answers and presses **Check**. An answer within 2% counts as right. **Show the solution** gives the worked steps. Once a lesson is solved, a box compares the hand result with what the app computes for the same barge.

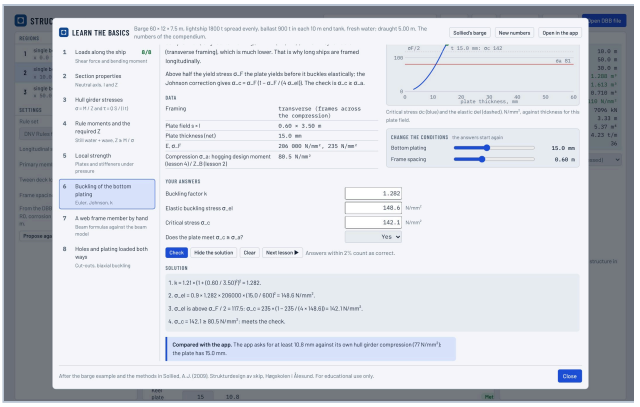
Lesson	By hand	Compared with the app
1. Loads along the ship	Draught, weight and buoyancy per metre, largest Q and M, hogging or sagging	Q and M from the numerical integration of the blocks
2. Section properties	Area, neutral axis, I, Z at deck and bottom (plates only)	The full section with keel plate, girders and stiffeners
3. Hull girder stresses	σ = M / Z at deck and bottom, S and τ at the neutral axis	The hull girder shear check
4. Rule moments	C_W, wave moments, design moment, required Z, the distribution factor	The required Z amidships
5. Local strength	k_a, plate thickness for pressure, minimum thickness, stiffener Z	The same plate and stiffener checks, same numbers
6. Buckling	k, σ_el, σ_c with the Johnson correction, the check	The plate buckling requirement
7. A web frame member	Line load, fixed-end and simply supported moments, end shear	The floor moment from the beam model, still water
8. Holes and two-way loading	Shear factor with a hole, τ, slenderness β, the interaction value	The flange check of the web frame

Sollied's barge gives the numbers of the compendium: 60 × 12 × 7.5 m, 1800 t lightship, 900 t ballast at each end, so Q = 600 t and M = 9000 t m hogging. **New numbers** makes another barge of the same kind for practice. **Open in the app** loads the barge, so the student can see its full structure, frames and checks. Every lesson has sliders for its conditions (ballast and tank length, plating, length and breadth at the same draught, frame and web frame spacing, hole size and frame stress). A change makes a new barge, recomputes the lesson and clears the answers. Show the solution fills every box with the right answer; changing an answer clears the marks until Check is pressed again.

A suggested use: lessons 1 to 4 in one session on longitudinal strength, 5 and 6 on local strength and buckling, 7 and 8 together with the Web frame tab.



Lesson 1, loads along the ship, with the sliders



Lesson 6, buckling, with the worked solution and the comparison with the app

Figure 11. Learn the basics.

12. Exercises for students

Each exercise uses an example or the student's own DBB design and asks for a short written answer with numbers from the app.

1. **Arrangement drives structure.** Open the PSV. Note the regions and the bulkhead positions. In the DBB editor, move one bulkhead block end by 3 m, save, and open again. Which regions change, and how do the steel weight per metre and the required section modulus change?
2. **Spacing.** For the tanker midship region, compare longitudinal spacings of 0.6, 0.7 and 0.8 m. Plot plate thickness, longitudinal profile and steel weight per metre against spacing. Which spacing is lightest, and why?
3. **Hull girder and local scantlings.** Raise the deck plating of a midship region by 2 mm. Watch the section modulus, the allowed stress in the longitudinals and the profiles they need. Explain the link.
4. **Check one plate by hand.** Pick a bottom plate. From its log, take p, s, the span and σ, compute t = 15.8 ka s √(p / (σ f1)) + tk on paper, and compare with the app.
5. **Read a web frame.** For the AHTS region 2, show the moment diagram for the full load and the ballast case. Where is the largest moment, which case gives it, and why? Sketch the deformed shape before you show it in the app.
6. **Loads that cancel.** Tick *Loads* for the tanker cargo region in full load. Explain why the double bottom moment is smaller in full load than in ballast.
7. **Build a load case.** Add a case *Half cargo, ballast in*: draught halfway between ballast and design, cargo tanks empty, ballast full. Does it govern any member? Propose again and compare the weights.
8. **Size a frame by hand.** Set every girder of one web frame to the smallest size, then step them up yourself until all are met. Compare your frame with *Propose this frame*: weight, usage and the governing case.
9. **Question an assumption.** Choose one row of section 9, explain how it affects the results, and test it in the app where you can (for example the head above the tank tops, or the tank test factor).

13. Symbols and units

Symbol	Meaning	Unit
L, B, D, T	Rule length, breadth, depth, draught	m
Cb	Block coefficient	–
x, y, z	Forward from the aft end, to port, up from the baseline	m
s	Longitudinal (stiffener) spacing	m
l	Span of a longitudinal = web frame spacing	m
S, b	Span and breadth carried by a primary member	m
t, tw	Plate thickness, web thickness	mm
tk	Corrosion addition	mm
ka	Aspect ratio factor of a plate field	–
f1	Material factor (1.0 for normal steel NV-NS)	–
p, q	Pressure, deck load	kN/m^2
h0	Depth below the waterline	m
pdp	Dynamic sea pressure term (Sec.4)	kN/m^2
ρ	Density	t/m^3
Z, W	Section modulus	cm^3 (elements), m^3 (hull girder)
I	Moment of inertia	cm^4, m^4
A, Az	Area, shear area	cm^2
Sy	First moment of area above the neutral axis	cm^3
be	Effective flange width	mm
N, Q, M	Axial force, shear force, bending moment	kN, kN, kNm
σ, τ, σe	Normal, shear, equivalent stress	N/mm^2
Usage	Largest stress / allowed stress	–

Line loads on the frame are pressures times the web frame spacing, in kN/m (equal to N/mm).

14. References and credits

References

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6. IACS UR S11, Longitudinal strength standard (wave moment and shear force formulas, as used in the DNV rules for ships).
7. Sollied, A.J. (2009). Strukturdesign av skip (compendium). Høgskolen i Ålesund. Rule moment and shear distributions, buckling with the Johnson correction, biaxial interaction, cut-outs in webs and the barge example.

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